

EDUCATION

University of Michigan Master of Engineering, Space Systems Engineering GPA: 3.79/4.0	Ann Arbor, MI, USA Aug 2024–Dec 2025
Vellore Institute of Technology (VIT) Bachelor of Technology, Mechanical Engineering GPA: 7.78/10	Vellore, India Jul 2019–Jul 2023

TECHNICAL SKILLS

Programming Languages: Python, LabVIEW (CLAD prep), MATLAB (learning), Verilog (learning), VHDL (learning)
Instrumentation & DAQ / Hardware: GPIB/IEEE-488, USB-serial, serial data logging, NI VISA (PyVISA, PyMeasure), Keithley DAQs and SMUs, TDK Lambda PSUs, High Voltage Switching Operations, Vacuum Chamber Operation (CTI-Cryogenics Hi-Vac; roughing pump operated at 1 mTorr)
FPGA, Plasma & Modeling Tools: SIMION, SRIM, Amptek DPPMCA, FPGA Design Workflow, AMD Vivado, MATLAB HDL Coder & DSP Toolbox, SPENVIS, SWMF & Magnetosphere-Ionosphere Models (MAGE, SWMF, HYPERs), Google Data Studio
Mechanical Engineering Tools: ANSYS (GUI/TUI): Fluent, Mechanical; PyANSYS, OptiSlang, OpenFOAM, SolidWorks, Fusion 360, AutoCAD, Autodesk Inventor, SpaceClaim, MIDO, Xflr5, RocketPy, OpenRocket

RESEARCH & ENGINEERING EXPERIENCE

Climate and Space Sciences and Engineering, University of Michigan Research Assistant, LVACCS Testing Rig Characterization - Validated a 1300 V hollow-cathode plasma-source test workflow by combining HV discharge-box FMEA, Python/PyVISA ignition control, PSU logging, and DAQ synchronization; cut manual steps 15→5 and achieved 98.6% synchronized data - Supported electrostatic regolith-lofting experiments using Retarding Potential Analyzer (RPA), emissive-probe, and Langmuir-probe diagnostics to compare plasma conditions across test states - Characterized remote ignition and post-run processing workflows for Spacecraft Charging Device test readiness, improving experimental repeatability and saving ~15 minutes of processing per run	Ann Arbor, MI, USA Feb 2026–Present
Climate and Space Sciences and Engineering, University of Michigan Research Assistant, Uranian Tropospheric Cloud Resolving Model Ran Uranian cloud-resolving model sensitivity studies on Great Lakes HPC, linking methane profile, microphysics, latent heating, gravity, and sedimentation effects to vertical-velocity and condensation diagnostics	Ann Arbor, MI, USA Sep 2025–Dec 2025
Solar and Heliospheric Research Group, University of Michigan Student Research Assistant, SSD Readout Chain - Designed an SSD readout roadmap around science-driven energy-resolution targets, integrating a Zmod ADC 1410 path with a Zynq-7000 architecture to evaluate a 1 MSPS→125 MSPS digitization upgrade - Verified MATLAB HDL Coder golden models against Vivado co-simulations and diagnosed 125 MHz post-synthesis timing limits, including Worst Negative Slack constraints in the Zynq SoC signal path	Ann Arbor, MI, USA Jan 2025–Dec 2025
Space Physics Research Lab, University of Michigan Summer Intern, TestBedz Spacecraft Qualification Platform Built a full-stack requirement-flowdown platform for TVAC, vibration, and EMI test submissions, matching spacecraft hardware profiles to facility limits across 6 test-type workflows	Ann Arbor, MI, USA May 2025–Jul 2025
L3Harris and University of Michigan Communications Lead, Uranian Orbiter and Probe Mission Study Led Uranian mission communications architecture trades and DSN/fault-tolerance requirement flowdown, managing 15 traceable requirements and supporting a 16% (\$450M) cost reduction through an orbiter-only configuration trade	Ann Arbor, MI, USA Sep 2024–May 2025

SELECTED TECHNICAL PROJECTS

Space Instrumentation Calibration & Ion-Optics Series <i>Graduate Course: SPACE 571 Supervisor: Prof. Stefano Livi</i> - Calibrated CEM detector response by mapping operating bias, pulse-height distributions, and amplifier gain behavior, connecting detector settings to usable signal quality and count-rate interpretation - Modeled ESA and ion-optics performance using lab energy-angle scans, SIMION, and SRIM to quantify transmission, angular acceptance, energy resolution, FWHM, and filter response	Jul 2025–Dec 2025
Spatial Ion-Electron Temperature Correlation <i>Graduate Course: SPACE 477 Supervisor: Prof. Xianzhe Jia</i> Analyzed ion-electron temperature structure in SWMF/HYPERS simulations, tracing magnetosheath ion distributions and reconnection-driven electron heating during simulated shock events	Jan 2025–Apr 2025

PROJECTS & LEADERSHIP

CANSAT 2022 (NASA & AAS) Ranked 7th worldwide out of 42 teams Built a 10 m tether-deployment, payload release, and camera-stabilization system using a DC motor, worm gear, torsion-spring latch, and dual-servo gimbal for controlled descent and fixed 45° imaging	Aug 2021–Jul 2022
Spaceport America Cup / IREC Placed 23rd globally and 5th in Asia-Pacific Launched a Mach 0.9 solid-motor rocket to 10,000 ft and developed an SRAD trajectory simulator using CFD-backed aerodynamic data for flight prediction	Apr 2020–Jul 2021
CANSAT 2021 (NASA & AAS) Placed 13th globally and 7th in Asia-Pacific Modeled maple-seed mono-wing payload descent using Blade Element Theory and transient CFD, with MATLAB telemetry software for real-time dual-payload deployment monitoring	Aug 2020–Jul 2021